National Guidelines for Primary Health Care Physicians

Hypertension
Prevention,Diagnosis, and Treatment

DirectorateofPublic Health
Non-communicableDiseases
Prevention and Control Department

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ContributorstotheNationalGuidelinesforHypertension:

- Dr. HilalBahjetAlSaffar, Consultant Cardiologist CABM,FRCP,FACC. International advisor, RCP London forIraq.,Headof ScientificCommittee, Iraqi Red Crescent Society.
- Dr. HaiderHusham Mohammed. General Medicine Specialist. Head of the Medical Section. Al-Imam Ali General Hospital. Baghdad Al-Risafa Health Directorate.
- A. Prof Dr Ghazi Farhan Haji, President of Iraqi Hypertension Society, Consultant Interventional Cardiologist, Baghdad College of Medicine
- Dr. Ala Sh. Ali, Consultant Nephrologist and Transplant Physician, FIBMS-Nephro, FASN, FACP, FRCP Nephrology and Renal Transplantation Centre, The Medical City -Baghdad
- Dr. MunaAttallahKhaleefah Ali, FICMS/ Community Medicine, Consultant, Director of Non Communicable Diseases (NCD) Prevention and Control Department, General Directorate of Public Health / MoH.
- Dr. Nada Abdul WahhabMousa/ PhD in Community Medicine, (NCD) Department, General Directorate of Public Health / MoH.
- Dr. Huda IbrahimAssal, Board Community Medicine, (NCD) Department General Directorate of Public Health / MoH.

ContributorstotheNationalGuidelinesforHypertension previousedition:

- Dr. HilalBahjetShawki Al Saffar, MD, FRCC, FACC Consultant Cardiologist, College of Medicine, Baghdad University, Head of Scientific Committee, Iraqi Cardiothoracic Society.
- Dr. Abbass Al Musawi, FRCP (Edin), Director of National Diabetes Centre, Al Mustansrya University.
- Dr. Hassan Yousif Al Najjar / DCC, FRCP (UK.IRE)/ Chairman of Cardiology, Iraqi Medical Specialization Board.
- Dr. Faris Abdul Kareem Khazzal, Assistant Professor, Al Kindi Medical College. AbbassKh. Al Wasity /General Medicine Specialist, Al Karkh General Hospital.
- Dr. KadhimHussienAbbood / Technical Directorate.
- Dr. Mohanned Mohammed Baqir/ FICMS/ Pathology/ Clinical Chemistry, Central Public Health Laboratory.
- Dr. MunaAttallahKhaleefah Ali, FICMS/ Community Medicine, Director of Non Communicable Section / MOH.
- Dr. Bushra Ibrahim Abd Al Lattif / MSc in Community Medicine , Primary Health Care Department, Directorate of Public Health , NCD Section.
- Dr. Nada Abdul WahhabMousa / PhD in Community Medicine, Primary Health Care Department, Directorate of Public Health , NCD Section

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Abbreviations:

ABPM	Ambulatory blood pressure monitoring
ACEI	Angiotensin converting enzyme inhibitor
ARBS	Angiotensin receptor blocker
BB	Beta Blocker
BMI	Body mass index
BP	Blood pressure
CCB	Calcium channel blocker
COPD	Chronic Obstructive Pulmonary Disease
DASH	Dietary approach to stop hypertension
DBP	Diastolic blood pressure
HBPM	Home blood pressure measurement
HF	Heart failure
HIC	high-income countries
HIIT	High intensity interval training
HMOD	Hypertension mediated organ damage
IHD	Ischemic heart disease
IRD	Inflammatory Rheumatic Disease
ISH	International Society of Hypertension
JNC	Joint National Committee
LMIC	low and middle income countries
LVH	Left Ventricular Hypertrophy
NICE	National Institute for Health and Care Excellence
OTC	Over the counter
SBP	Systolic blood pressure

Introduction

Hypertension is oneofthemost common worldwidehealth problems,as itis estimated that over20% ofall adults across theworld havehypertension. Developingcountries are experiencingdramaticchanges in health needs. Theincreasingprevalenceof chronic diseases as hypertension isincreasing globally creatingapublichealth challenge and creating an adverse impact on cardiovascular morbidity and mortality, irrespective of income.

Raised BP remains the leading cause of death globally accounting for 10.4 million deaths per year. BP trends show a clear shift of the highest BPs from high-income to low-income regions.The large disparities in the regional burden of hypertension are accompanied by low levels of awareness, treatment and control rates in low and middleincome countries (LMIC).Theincidenceof hypertension is growing amongwomen and adolescents as well as theolder adults.

Accordingto the2015Iraqi national surveyfor chronicdiseaserisk factors,35.6%ofthe Iraqi adultspopulation (18+ years) have elevated blood pressure.

General principles:

- **Hypertensionandcardiovascular diseases:**Therelationship betweenBPandriskof CVDeventsiscontinuous,consistent,andindependentofotherriskfactors.The highertheBP,thegreateristhechanceofheartattack,heartfailure,stroke,and kidney disease.Forindividuals40–70yearsofage,eachincrementof20mmHgin systolicBP(SBP)or10mmHg indiastolicBP(DBP)doublestheriskofCVDacross the entireBPrange

- **Benefitoflowering BP:**In clinical trials, antihypertensive therapy has been associated with reductions in incidence of stroke 35–40%, myocardialinfarction 20–25%, and heart failuremorethan 50%

Definition of hypertension:

Based on the WHO/ISH recommendations, and the JNC8,NICE guidelines, it is recommended that hypertension be diagnosed in adults aged 18 years or older when aperson's systolic blood pressure (SBP) in the office or clinic is ≥140 mm Hg and/or the diastolic bloodpressure (DBP) is ≥90 mm Hg following repeated examinations.

Usually 2–3 office visits at 1–4-week intervals (depending on the BP level) are required to confirm the diagnosis of hypertension.

The diagnosis might be made on a single visit, if BP is ≥180/110 mm Hg and there is evidence of cardiovasculardisease (CVD).

Classification of hypertension:

According to ISH guideline, classification of hypertension is presented in table(1).

Table (1): Classification of Hypertension Based on Office Blood Pressure (BP) Measurement			
Category	**Systolic BP(mm Hg)**		**Diastolic BP(mm Hg)**
Normal BP	<130	And	<85
High-normal BP	130-139	and/or	85-89
Grade 1 Hypertension	140-159	and/or	90-99
Grade 2 Hypertension	≥160	and/or	≥100

Source:International society of hypertension global hypertension practice guidelines (2020)

High-normal BP is intended to identify individuals who could benefit from lifestyle interventions and who would receive pharmacological treatment if compelling indications are present.

Ambulatory and home BP measurements are used to confirm the diagnosis of hypertension in people with a clinic blood pressure of 140/90 mmHg or higher and ABPM daytime average or HBPM average of 135/85 mmHg or higher(table 2, annex1).

Table (2): Criteria for Hypertension Based on Office-, Ambulatory (ABPM), and Home Blood Pressure (HBPM) MeasurementSBP/DBP mm Hg	
	SBP/DBP, mm Hg
Office BP	≥140 and/or ≥90
ABPM	
24 -h average	≥130 and/or ≥80
Day time (or awake) average	≥135 and/or ≥85
Night time (or asleep) average	≥120 and/or ≥70
HBPM	≥135 and/or ≥85

Source:International society of hypertension global hypertension practiceguideline (2020)

Types of hypertension:

Essential hypertension:

That represents over90%ofthecases.It is diagnosed in the absenceofan identifiable cause.However, the following conditions increasethelikelihood ofits occurrence(Table3).

Table(3): conditions increase the likelihood of essential hypertension
Obesity Diabetes/ insulin resistance High salt intake High fat intake Morethan moderateintakeof alcohol Inadequate intake of fruits, vegetables (low potassium) Sedentarylifestyle Stress Familyhistoryofhypertension, age,gender and race (non modifiable risk).

Secondaryhypertension:

That represents less than10%of all cases.It is defined as hypertension resultingfrom anunderlyingidentifiablecause (Table4). Early diagnosis of secondary hypertension and the institution of appropriate targeted treatment have the potential to cure hypertension in some patients or improve BP control/reduce the number of prescribed antihypertensive medications in others.

Table(4): Identifiable causes ofsecondary hypertension

Sourceorcategory ofcause	Possiblecauses
Renaldiseases	Renalparenchymaldisease Polycystickidney Urinarytractobstruction Rennin-producingtumor Liddlesyndrome
Renovascularhypertension	Renalarterystenosis Connectivetissuedisease Glomerulonephritis
Vascular	Coarctationofaorta Vasculitis/polycythemia Collagen vasculardisease
Hormoneand steroidintake	Oralcontraceptives Estrogenreplacement therapy Oraland Depotcontraceptives,
Adrenal	Primaryaldosteronism Cushingsyndrome Pheochromocytoma Congenitaladrenalhyperplasia
Otherendocrinedisorders	Hyperthyroidismand hypothyroidism Hyperparathyroidism Acromegaly

Neurogenic	Braintumor Lesionsofbrainstemorhypothalamus Raisedintracranialpressure
Prohypertensivesubstances	Adrenergicmedication, nasal decongestants Nonsteroidalanti-inflammatorydrugs Anti-depressants(tricyclic,MAOI), Alcohol, CyclosporineandTacrolimus,
Other	Pregnancy Hypercalcemia Sleep Apnea

White Coat and Masked Hypertension

White coat hypertension refers to those who have elevated BP only in the office (nonelevated ambulatory or home BP).These subjects are at intermediate cardiovascular risk between normotensives andsustained hypertensives. The diagnosis needs confirmation with repeated office and out-of-officeblood pressure measurements.
If their total cardiovascular risk is low and there is no hypertension-mediated organ damage (HMOD), drug treatment may not be prescribed. However, they should be followed with lifestyle modification, as they may develop sustained hypertension requiring drug treatment.

Masked hypertension refers to those who have nonelevated BP in the office but elevated BP out of the office (ambulatory or home).Those patients are at similar risk of cardiovascular events as sustained hypertensives. The diagnosis needs confirmation with repeated office and out-of-office measurements. Masked hypertension may require drug treatment aiming to normalize out-of-office BP.

Malignant or accelerated hypertension

It is characterizedby increase in blood pressure to 180/120 mmHg or higher.This clinical entity may complicate hypertensionof any etiology and characterized byaccelerated microvascular damage with necrosis in the wall of small arteries and arterioles and intravascular thrombosis.
The diagnoses is made clinically by a high BP and rapidly progressive end organ damage such as retinopathy (grade 3 or 4), renal dysfunction (especially proteinuria) and/or hypertensive encephalopathy.If left untreated, death will occur within few months.

Seasonal BP Variation

BP exhibits seasonal variation with lower levels at higher temperatures and higher at lower temperatures. Similar changes occur in people traveling from places with cold to hot temperature, or the reverse.

Evaluation of patients with confirmed hypertension

Evaluation ofpatients with confirmed hypertension has threeobjectives, allofwhich contribute to risk stratification of hypertensive patients that may affect prognosis and guide management. These objectives are as follows:

(1) To assess lifestyle and identify other cardiovascular risk factors or concomitant disorders(table5).
(2) To reveal identifiable causes of high BP (secondary hypertension) (see table 4).
(3) To assess the presence or absence of Hypertension mediated organ damage (HMOD) that refers to the structural or functional alteration of the arterial vasculature and/or the organs it supplies (table 6).

Table (5): CardiovascularRiskFactors:
• HypertensionBP≥140/90 • Agemen >55years, women >65years • Cigarettesmoking • Overweight orObesity − Bodymass index≥25 Kg/M^2 − Abdominal obesity(waist circumferenceof>102cm formen, >88cm. forwomen) • Physical inactivity • Dyslipidemia − Total Cholesterol > 200 mg/dl. (5.2 mmol/L) − HDL in men <40 mg/dl (<1 mmol/L); and in women < 45 mg/dl. (<1.2 mmol/L) − Triglyerides> 150 mg/dl. (1.7 mmol.L) • Diabetes Mellitus or impaired glucose tolerance • Renal dysfunction − Microalbuminuria (urine albumin >300 mg/dl.) − Estimated GFR <60ml/min • Family history of premature cardiovascular disease (man <55 years or woman <65 years) in first degree relative

Table (6): Hypertension Mediated Organ Damage (HMOD)
• Heart - Left ventricular hypertrophy - Ischemic Heart Disease - Heart failure
• Brain - Stroke or transient ischemic attack
• Arteries - Carotid arteries atherosclerotic plaque /stenosis - Aortic disease - Lower extremity arteries
• Eye - Retinopathy - Papilledema
• Kidney - Proteinuria - Urine sediments - and/or low GFR

Assessment:

History:

To adequatelyevaluate accompanyingrisk factors, possibletreatablecauses, and hypertension-mediated organ damage (HMOD), acareful andcomprehensive clinical historyand physical examination is essential. Thehistoryshould include:

- Familyhistoryofthe following:
  - hypertension
  - diabetes
  - dyslipidaemia
  - chronic heart disease
  - stroke (CVA)
  - renal disease.
- Duration and previous levels of blood pressure, and results and side effects of previous antihypertensive therapy.
- Past historyor current symptoms ofthe following,as well as information about the drugs used to treat theseconditions:
  - Chronic heart disease and heart failure
  - Cerebrovascular disease
  - Peripheral vascular disease
  - Diabetes
  - Gout
  - Dyslipidemia
  - Bronchospam /chronic respiratory diseases
  - Sexual dysfunction
  - Renal disease
  - Othersignificant illnesses
- Symptoms suggestiveofsecondarycauses ofhypertension, such as:
  - Loud nocturnal snoring or interrupted respirations
  - Hypertension in childhood or adolescence
  - Palpitations
  - Frequent episodes of sweating or tremor
  - Abdominal and back pain
  - Evidenceofrenal diseaseorinsufficiency
- Careful assessment oflifestyleand factors including:
  - Dietary intake of fat
  - Dietary intake of salt (sodium)
  - Possible use of alcohol
  - Historyofsmoking

- Frequency of and type of regular physical activity
- History of weight gain since early adult life (as useful index of excess body fat)
- Past historyofdruguseoruseofsubstances that can raiseblood pressure, such as:NSAIDs or over the counter OTC drugs
  hormones, steroids and otherprohypertensivedrugs and substances
- Personal, psychological and environmental factorsthat could influencethecourse and outcomeof antihypertensive care, including familysituation, work environment and educational background.

Physicalexamination:

Points to be considered:

- General signs suggestive of secondary causes and other risk factors like :
  - Characteristic faces of Cushing syndrome.
  - Signs of acromegaly
- Presence of other risk factors: central obesity, tendon xanthomas (dyslipedemia)
- Measurement of height and weight and waist circumference, with calculation of BodyMass Index (BMI).
- Cardiovascular system:
  - Apical heave, (LVH)
  - Loud S2 or S4, JVP (evidence of heart failure)
  - Arterial disease (carotid, peripheral, renal)
  - Radial-femoral delay of pulse (coarctation of aorta)
  - Dependent edema
- Lungs:
  - Basal crackles
  - Wheezes
- Abdomen:
  - Palpable kidney (polycystic kidney)
  - Other masses(abdominal aortic aneurysm)
  - Bruit (renal artery stenosis)
- Fundoscopy:
  - Arteries: tortuosity, thickening, arterio-venous ripping
  - Background: haemorrhage, exudates, diabetic retinopathy
  - Discs: papilledema
- Nervous system:
  - Evidence of previous neurological disease
- Endocrine:
  - Evidence of Cushing's syndrome, thyroid disease

Laboratory Tests andOtherDiagnosticprocedures:

Routinelaboratorytests recommended beforeinitiatingtherapyinclude(all patients):

- Urinalysis for blood and protein
- Hematocrit CBC
- Chemical profile: Blood glucose, urea, creatinine (or the corresponding estimated glomerular filtration rate (e GFR), potassium, calcium, uric acid.
- Lipid profile: total cholesterol, (TC) high density lipoprotein cholesterol (HDL), low- density lipoprotein cholesterol (LDL), and triglycerides (TG).
- TwelveleadECG.

Tests forselected patients:

- CXR to detect cardiomegaly, heart failure and coarctation of aorta.
- Ambulatory blood pressure monitoring to assess borderline hypertension or white coat hypertension.
- Echocardiography to detect LVH.
- Nitrouritic peptide for heart failure detection.
- Abdominal ultrasound to detect renal disease, aortic aneurysm.
- Thyroid function test
- Urinary catecholamine or VMA (and plasma catecholamine)(mandilic acid) to detect pheochromocytoma.
- Urinary cortisol and dexamethasone suppression test to detect Cushing syndrome.
- Plasma renin activityandaldosteroneto detect primaryaldosteronism.

CVD Riskprediction:

Cardiovascularriskcanbeexpressedasthepercentagechanceofanindividualexperiencing acardiovasculareventoverapre-definedperiodoftime,usually thenext10years.Itmainly dependsonthepresenceof CVDriskfactorssuchassmoking,averagebloodpressure, BMI,cholesterollevels,age,andpresenceorabsenceofdiabetes.

The chart below can be used to predict the cardiovascular disease risk over the next 10 years in areas that have access to total cholesterol measurements (Fig1) or those with no access to cholesterol measurement with BMI measurement (Fig2):

Figure (1):WHO Risk Prediction Chart with Cholesterol Measurement with BMI

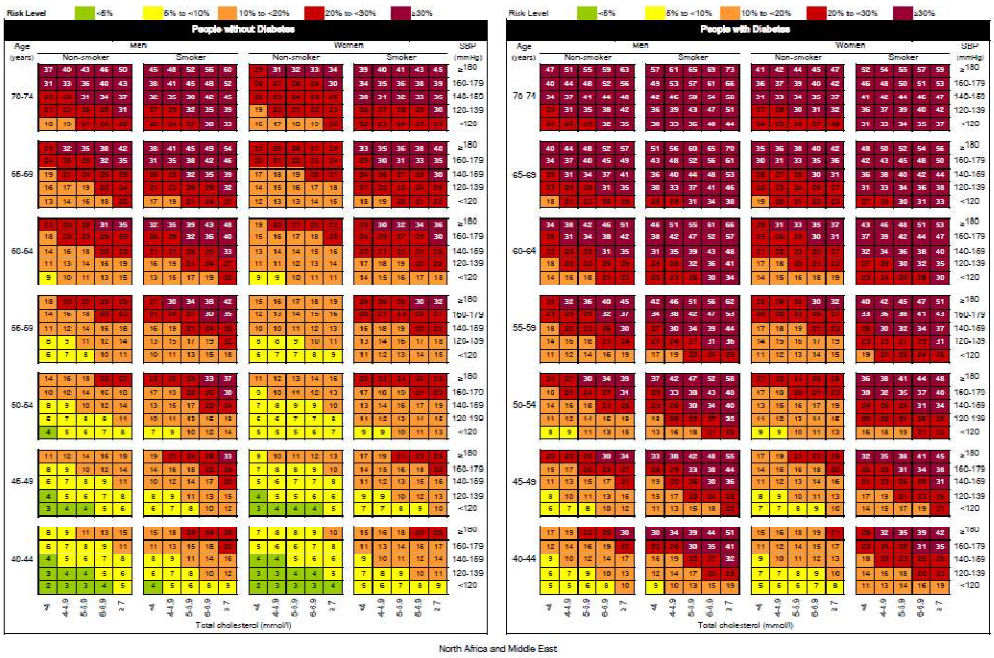
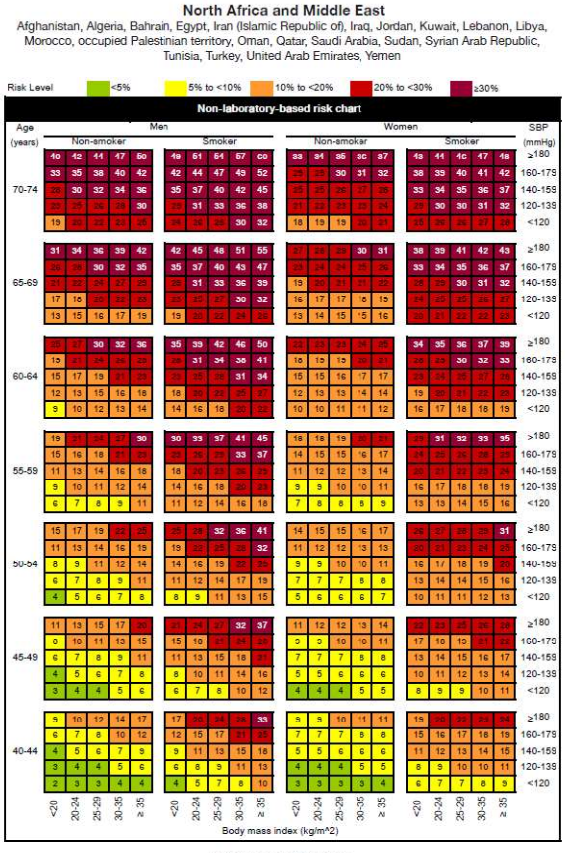


Figure (2): WHO Risk Prediction chart WITHOUT Cholesterol Measurement with BMI



Management Strategy of hypertension

1. Classification ofthedegreeofhypertension ofthepatient based on Table1.
2. Determiningtheoverall risk profileofthepatient based on theWHO Risk Prediction charts (fig 1,2).
3. Determiningthespecifictherapeuticgoals fortheindividual patient anddevelop aplan to lowerBPandreducethe hypertension mediated organ damage(HMOD) including:
   - Lifestylemodification to decrease risk factors.
   - Pharmacological treatment (annex2).
   - Monitoringblood pressure and contributory risk factors.

Lifestyle Modifications counseling (annex3A):

Lifestylemodifications mayreducetheincidenceofhypertension in high-normal blood pressure, decreasethedose and/orthenumberofdrugs needed inpatients with established hypertension,and reducecardiovascular risk.

Lifestylemodifications shownto lowerBPinclude:

1. Weight reduction in thoseindividuals who areoverweight orobese. Particularly abdominal obesity.
2. Salt reduction, healthy diet (annex2) and healthy drinks.
3. Practicingregular physical activity
4. Cessation ofsmoking
5. Reduction of alcohol intakeif alcohol taken

Potassiumsupplementation:

Someofthe advantagesofalowersodium intakemaybe related to its tendencyto increase bodypotassium content, byacoincidental increasein dietarypotassium intake and bya decreasein potassium wastageifdiuretics arebeingused.potassium supplements aretoo costlyand potentiallyhazardous for routineusebynormokalemichypertensivepersons. The best sourceis increasedconsumption of fruits and vegetables.

Otherdietary constituents:

Loweringblood pressurehas been notedwith diet rich in fruits and vegetables that couldreflect increases in fiber,potassium, orotheringredients, and with high doses ofomega-3 fattyacids from fish oil and supplements ofvitamin C. Caffeineacutelybuttransientlyraises blood pressure. High consumption of alcohol raises blood pressure.

Relaxationtechniques:

Most studies ofvarious cognitive-behavioral therapies haveshown transient but not sustained lowering ofblood pressure.

Combinedtherapies:

When several life-stylemodifications are combined, additional effects mayaccrue.

Therapeutic goal:

Despite the fact that areduction ofBP by20/10 mm Hg is associated with a 50%difference in cardiovascular risk, thetherapeutic goal oflong-term management of hypertension is to maintain ablood pressureofless than 140/90; that is, BOTH asystolicpressureofless than 140 and adiastolic pressureofless than 90 for as long as thepatient lives.

Becauseofthehigher risk of cardiovascularcomplications in thosewith current diabetes (TypeIorII)or renal disease, thegoal of management in thesepatients is blood pressureofless than 130/80.

PharmacologicTreatment(annex IV):

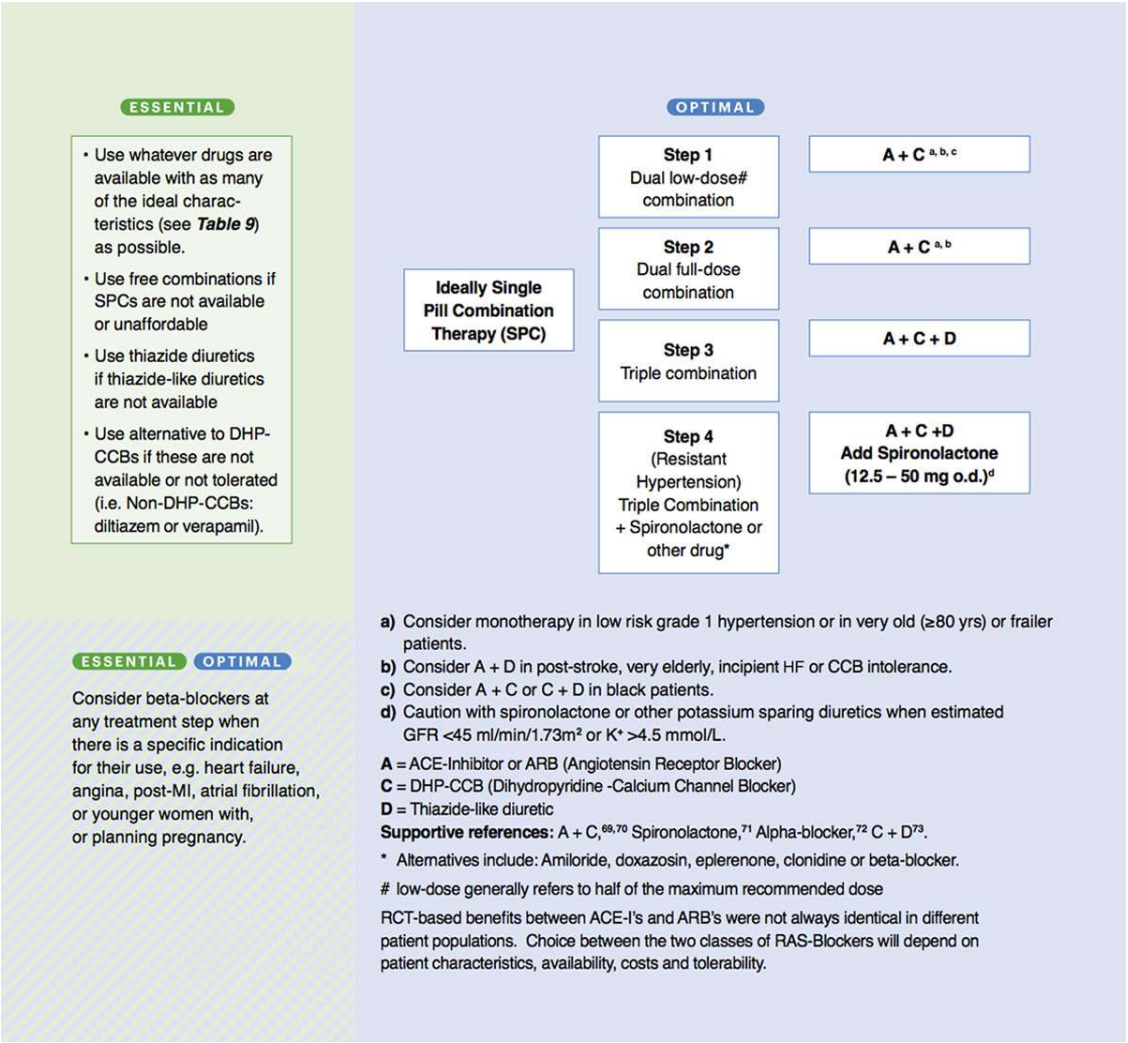
The main classes of antihypertensive medications:

1. Thiazide and thiazide-like diuretics
2. Angiotensin converting enzyme (ACE) inhibitors
3. Angiotensin receptor blockers (ARB)
4. Calcium channel blockers (CCB)
5. Others (β blockers, aldosterone antagonists (ALDOANT), α blockers Vasodialators, centrally acting drugs….)

Thiazidediuretics havebeen proven to be efficientin achievingblood pressure control and preventingthecardiovascular complications ofhypertension with few sideeffects. Theyalso enhancethe antihypertensive efficacyofmultipledrug regimens. Becauseofthis, theyare used as initial therapyformost patients with hypertension, either aloneorincombinationwith oneoftheotherclasses which havebeen demonstrated to bebeneficialin randomized controlled outcometrials. Othermorepotent loopdiuretics such as furosemide, bumetanide have few advantages overthiazide except in patients with renal impairment. Ifadrugis nottolerated oris contraindicated, then oneoftheother classes proven to reducecardiovascular events should beused instead.

Theother classes of antihypertensivedrugs include angiotensin converting enzymeinhibitors (ACEI), angiotensin receptorblockers (ARB), beta-blockers (BB), calciumchannel blockers (CCB), aldosterone antagonists (ALDOANT)should especiallybeusedforthetreatment of hypertension ifthere areco-existingclinical conditions such as thoselisted below in fig 3 and table7. Their effectiveness in reducingthemorbidityofthese co-existingconditions as well as in reducingblood pressurecreates a compellingindication fortheiruse.

Figure (3): ISH core drug treatment strategy



Source:International society of hypertension global hypertension practice guidelines (2020).

Table(7): Compelling IndicationsforIndividual Drug Classes

Compelling Indication*	RecommendedDrugs+					
	Diuretic	BB	ACEI	ARB	CCB	ALDO ANT
Heartfailure	Yes	Yes	Yes	Yes		Yes
Postmyocardial infarction		Yes	Yes			Yes
Highcoronary diseaserisk	Yes	Yes	Yes		Yes	
Diabetes	Yes	Yes	Yes	Yes	Yes	
Chronickidney Disease			Yes	Yes		
Recurrentstroke Prevention	Yes		Yes			

*Compelling indicationsfor antihypertensivedrugsarebasedon benefitsfromoutcomestudiesor existing clinical guidelines;thecompelling indication ismanagedin parallelwiththeBP.

+ Drug abbreviations:ACEI, angiotensin converting enzymeinhibitor;ARB,angiotensin receptor blocker;Aldo ANT, aldosterone antagonist;BB,beta-blocker;CCB,calciumchannelblocker.

Achieving BloodPressureControl inIndividual Patients:

Most patients who arehypertensivewill requiretwo ormore antihypertensivemedications to achievetheirBPgoals. Addition ofasecond drug from adifferent class should beinitiated when useofasingledrugin adequatedosesfailsto achievetheBPgoal.When BPis more than 20/10 mmHgabovegoal,consideration should begiven to initiatingtherapywith two drugs,either as separateprescriptions orin fixed-dose combinations.

Theinitiation ofdrugtherapywith morethan oneagent mayincreasethelikelihood of achievingtheBPgoal in amoretimelyfashion, but particularcaution is advised in those at risk fororthostatichypotension, such as patientswith diabetes, autonomicdysfunction, and someolderpersons. Useofgenericandfixed combination drugs should be considered to reduceprescription costs.

Key points in drug therapy

- In unselected group of hypertensive population, no one class of agents is any more effective at lowering BP than another.
- Begin with the lowest possible dose of the particular agent so as to reduce BP by 5-10 mmHg at a time.Too rapid BP lowering may lead to undue side effects (fatigue, lethargy and postural dizziness) and this may affect the compliance of the patient.
- If this is not successful at achieving the BP goal, then increase the dose gradually provided that it is well tolerated.
- Using low doses of two drugs that act by different mechanisms, allows better BP control with least side effects. Examples of useful combinations:
    - Thiazide diuretic can be used in combination with all antihypertensive drugs
    - ACEI + CCB
    - Dihydropyridine CCB + BB
    - ACEI + ARB .
- Some combinations may have additive ADVERSE effects, such as BB + Verapamil or diltiazem (can lead to bradycardia, cardiac depression and heart block)
- Change to a different class of antihypertensive drug if there is little response, poor tolerability or side effect. This should be done cautiously with further consultation, especially if discontinuing B-Blocker medication.
- Remember that the presence of certain clinical conditions in hypertensive patient (IHD, HF, renal disease, dyslipidemia, prostatic hypertrophy, COBD) may necessitate the use of certain classes of antihypertensive drugs.
- Using drugs with long duration of action is preferred to improve patient compliance as to be taken in a single dose daily.
- The best time to take the drug with long duration of action is as early as possible in the morning to cover the early morning surge of BP.
- The patient should know about his medication, dosing, mode of actions and possible side effects.
- When possible, to improve compliance with the management program, other staff

members such as nurses should be trained to counsel the patient regarding the medications, dosing, and possible side effects

- The importance of regular visits to monitor the effectiveness of therapy and lifestyle changes should be emphasized.
- Treatment should be affordable and/or cost-effective relative to other agents.
- Treatments should be well-tolerated
- If the patient is already on another medication regimen, blood pressure is controlledto the target level, and the medications the patient is taking are accessibleand affordable, there is no reason to change the regimen.
- Antihypertensive drugs can have both favorable or unfavorable effects on other possible co- morbidities This can influence the indications and contraindications of specific antihypertensive drugs in these selected patients (table8).
- Pregnant women and women of childbearing age not on effective contraception should not be given ACE inhibitors, ARBs, or thiazide/thiazide-like diuretics; CCBs should be used. If not controlled with intensification dose of medication, refer to specialist.
- Beta blockers are not recommended as first-line therapy. If a heart attack has been diagnosed within the previous three years, or there is atrial fibrillation or heart failure, then a beta blocker should be added to the starting dose of antihypertensive medication. Patients with angina may also benefit from treatment with a beta blocker.

Other treatment considerations

- If there is a prior heart attack or stroke, or the person is otherwise at high risk of CVD, start a statin at the same time as starting antihypertensive medication. (Statins should not be used in women who are or who may become pregnant.)
- If there is a prior heart attack or ischemic stroke, start low-dose aspirin.

Table(8): Indications andcontra indications ofantihypertensivedrugs

-- Indications and Contraindications for the Use of Antihypertensive Drugs

Class of Drug	Indications	Contraindications
Diuretics	Heart failure	Gout
	Advanced age	
	Systolic hypertension	
Beta blockers	Angina or previous myocardial infarction	Asthma or chronic obstructive pulmonary disease
	Heart failure	Heart block
	Tachyarrhythmias	
	Migraine	
Alpha blockers	Prostatic hypertrophy	Incipient heart failure
Calcium channel blockers	Advanced age	Heart block (verapamil, diltiazem)
	Systolic hypertension	
	Cyclosporine-induced hypertension	
Angiotensin-converting enzyme inhibitors	Heart failure or left ventricular dysfunction	Pregnancy
	Previous myocardial infarction	Bilateral renal artery stenosis
	Diabetic or other nephropathy or proteinuria	Hyperkalemia
Angiotensin receptor blockers	Angiotensin-converting enzyme inhibitor–associated cough	Pregnancy
	Diabetic or other nephropathy	Bilateral renal artery stenosis
	Congestive heart failure	Hyperkalemia

Source:.Libby:Braunwald'sHeart Disease: ATextbookofCardiovascularMedicine,8thed.

Non-compliance with drug treatment

Failure of compliance with drug treatment is a major problem in all hypertensive patients which may reach up to 60%, compliance of hypertensive patients should be assessed, may result from many causes:

- Complex drug regimen.
- Nature of the doctor –patient relationship.
- Frequent dosing schedules.
- Side effects.
- Cost of medication.
- Resistant hypertension which need multi drug regimen and more side effects.

Strategies to improve compliance to treatment:

- Reducing polypharmacy – use of fixed drug combination as linking adherence behavior with daily habits.
- Reminder packaging of medications.
- Empowerment-based counseling for self-monitoring.
- multidisciplinary healthcare team approach (ie, pharmacists) to improvemonitoring for adherence.

Adherence to treatment is critical for blood pressure control. If antihypertensive medication is being prescribed, the following are critical to ensuring adherence:

- Explain the difference between medicines for long-term control (for example, of blood pressure) and medicines for quick relief (such as for headaches).
- Explain the reason for prescribing the medicine(s).
- Explain the diagnosis of hypertension.
- Discuss the asymptomatic nature of hypertension and explain that medications must be taken even if there are no symptoms.
- Inform patient of the complications of untreated hypertension, including stroke, heart attack, kidney failure.
- Explain the disability and economic and family burden these preventable complications cause.
- Teach the patient how to take the medications at home.
- Show the patient the appropriate dose.
- Explain how many times a day the patient should take the medication and at what time, and adopt the following simple steps to help them to adhere to the guidelines
- Label and package the tablets.
- Check the patient's understanding before the patient leaves the health centre.
- Wherever possible, use once-daily dosages of all medications, to be given at the same time each day.

Specialconsiderationsin Management of Hypertension

IschemicHeartDisease

Ischemicheart disease(IHD)is themost common form of HMOD associated with hypertension. In patients with hypertension and stableanginapectoris, the first drugof choiceis usuallya BB; alternatively, long-actingCCBs.In patients with acute coronarysyndromes (unstable anginaormyocardial infarction), hypertension should be treated initiallywithBBsand ACEIs, with addition ofotherdrugs as needed forBPcontrol. In patients with postmyocardial infarction, ACEIs, BBs,and aldosterone antagonists have proven to bemost beneficial.

HeartFailure

Heart failure(HF),in the form ofsystolicordiastolicventriculardysfunction, results primarilyfrom systolichypertension andIHD. Inasymptomatic individuals with demonstrableventriculardysfunction, ACEIs andBBs arerecommended. Forthosewith symptomaticventriculardysfunction or end-stageheart disease, ACEIs,BBs, ARBs and aldosteroneblockers arerecommendedalongwith loop diuretics.

Diabetes withHypertension

Combinations oftwo ormoredrugs areusuallyneeded to achievethetarget goal of<130/80 mmHg.ACEIorARBbased treatments favorablyaffecttheprogression ofdiabeticnephropathyand reducealbuminuria, andARBs havebeen shown to

reduceprogression tomacroalbuminuria. Thiazidediuretics, BBs,ACEIs, ARBs, and CCBs havebeen shown to bebeneficial in reducingCVD and strokeincidencein patients with diabetes.

ChronicKidney Disease

In peoplewith chronickidneydisease (CKD), as defined byeither
(1)Reducedexcretoryfunction with an estimated GFRbelow 60 ml/min per1.73 m^2
OR
(2)Thepresenceofalbuminuria (>300 mg/dayor200 mgalbumin/g creatinine). Therapeutic goals areto slow deterioration of renal function and prevent CVD. Hypertension appears in themajorityofthesepatients, and theyshould receiveaggressiveBPmanagement, often with threeormoredrugs to reach target BPvalues of<130/80 mmHg.

ACEIsand ARBs havedemonstrated favorableeffects on theprogressionofdiabetic andnon-diabeticrenal disease. A limited risein serumcreatinineof as much as35 percentabove baselinewith ACEIs orARBs is acceptableand is not a reason to withhold treatment unless hyperkalemiadevelops.With advanced renal disease(estimated GFR <30 ml/min 1.73 m2), increasingdoses ofloop diuretics are usuallyneeded in combination with otherdrugclasses.

CerebrovascularDisease

The risks and benefits ofacuteloweringofBPduringanacutestroke arestill unclear; control ofBPatintermediatelevels (approximately160/100 mmHg)is appropriateuntil the condition has stabilized orimproved. Recurrent stroke
ratesarelowered bythecombination of an ACEI and thiazide-typediuretic.

Obesity andthemetabolicsyndrome

Obesity(BMI>30 kg/m2)is an increasinglyprevalent risk factor forthedevelopment of hypertension and CVD. Metabolicsyndromedefined as thepresenceof and increased waist circumference (waist circumference>102cm. in men or>88 cm. in women)with two or

moreofthe followingconditions: glucoseintolerance (fasting glucose>100 mg/dL),BP >130/80 mmHg, high triglycerides (>150 mg/dL), orlow HDL(<40 mg/dLin men or<50 mg/dLin women).

Intensivelifestylemodification should bepursuedin all individuals with themetabolic syndrome,and appropriatedrugtherapyshould beinstituted for each ofitscomponents as indicated.

Leftventricularhypertrophy

Left ventricularhypertrophy(LVH)is an independent risk factorthat increases the risk of subsequent CVD. Regression ofLVH occurs with aggressiveBPmanagement, including weight loss, sodium restriction, and treatment with all classes of antihypertensiveagents except thedirect vasodilators hydralazine, and minoxidil.

Peripheral arterial disease

Peripheral arterial disease (PAD)is equivalent in risk tolHD. Anyclass of anti-hypertensivedrugs can beused in most PAD patients. Other risk factors should be managed aggressively, and aspirin should beused.

Hypertensioninolder persons

Hypertension occurs inmorethan two-thirds ofindividuals afterage65.This is also the population with thelowest rates ofBPcontrol. Treatment recommendations forolderpeople with hypertension, includingthosewho haveisolated systolichypertension,should follow the sameprinciples outlinedforthegeneral careofhypertension.In manyindividuals, lower initial drugdoses maybeindicated to avoid symptoms; however, standarddoses and multiple drugsareneeded in themajorityofolderpeopleto reachappropriateBPtargets. Thiazde diuretics and CCBarethepreferred classes ofdrugs to beused inelderlypatients.

Postural hypotension

A decreasein standingSBP>10 mmHg, whenassociated with dizziness orfainting, is more frequent in olderpatientswith systolichypertension, diabetes, and thosetakingdiuretics, venodilators(e.g., nitrates, alpha-blockers, and sildenafil likedrugs),and somepsychotropic drugs.BPin theseindividuals should also bemonitored in theupright position. Caution should beused to avoid volumedepletion and excessivelyrapid dosetitration of antihypertensivedrugs.

Hypertensioninwomen

Oral contraceptives mayincreaseBP, and therisk ofhypertension increaseswith duration of use. Women takingoral contraceptives should havetheirBPchecked regularly. Developmentofhypertension is areason to considerother forms of contraception.In contrast, menopausal hormonetherapydoes not raiseBP.
Women with hypertension who becomepregnant should be followed carefullybecauseof increased risks to motherand fetus. Methyldopa,BBs,and vasodilators arepreferred medications forthesafetyofthefetus. ACEI andARBs should

not beusedduringpregnancy becauseofthepotential for fetal defectsand should be avoided in womenwho arelikelyto becomepregnant.

Hypertension in Pregnancy

Hypertension in pregnancy is a condition affecting 5%–10% of pregnancies worldwide. Maternal risks include placental abruption, stroke, multiple organ failure (liver, kidney), disseminated vascular coagulation. Fetal risks include intrauterine growth retardation, preterm birth, intrauterine death.

Pre-eclampsia, which occurs afterthe20th weekofpregnancy, is characterized bynew-onset orworseninghypertension, albuminuria, and hyperuricemia, sometimes with coagulation abnormalities.In somepatients, preeclampsiamaydevelop into ahypertensiveurgencyor emergencyand mayrequirehospitalization, intensivemonitoring, earlyfetal delivery,and parenteral antihypertensive

Hypertension and Chronic Obstructive Pulmonary Disease (COPD)

Hypertension is the most frequent comorbidity in patients with COPD. BP should be lowered if ≥140/90 mm Hg and treated to a target <130/80 mm Hg (<140/80 in elderly patients).Lifestyle changes (smoking cessation) are mandatory.Environmental (air) pollution should be considered and avoided if possible.

The treatment strategy should include an angiotensin AT1-receptor blocker (ARB) and CCB and/or diuretic, while beta blockers (s1-receptor selective) may be used in selected patients (eg, CAD, HF).Additional cardiovascular risk factors should be managed according to cardiovascular risk profile.

Hypertension and Psychiatric Diseases

The prevalence of hypertension is increased in patients with psychiatric disorders and in particular depression.According to guidelines, psychosocial stress and major psychiatric disorders increase the cardiovascular risk.

Depression has been associated with cardiovascular morbidity and mortality, suggesting the importance of BP control.

Hypertension and Inflammatory Rheumatic Diseases (IRD)

IRD (rheumatoid arthritis, psoriasis-arthritis, etc) are associated with an increased prevalence of hypertension under diagnosed and poorly controlled.IRD show an increase in cardiovascular risk only partially related to cardiovascular risk factors. Rheumatoid arthritis is predominant among IRD.The presence of IRD should increase 1 step of cardiovascular risk.

Hypertensive urgencies and emergencies

Emergency:
Patients with marked BP elevations and acute HMOD (e.g., encephalopathy, myocardial infarction, unstable angina, pulmonary edema, eclampsia, intra cranial hemorrhage , head trauma, life-threatening arterial bleeding, or aortic dissection) require hospitalization and parenteral drug therapy.

Urgency:
Patients with markedly elevated BP but without acute HMOD usually do not require hospitalization, but they should receive immediate combination oral antihypertensive therapy. They should be carefully evaluated and monitored for hypertension-induced heart and kidney damage and for identifiable causes of hypertension.

ResistantHypertension

Resistant hypertension is the failureto reachgoalBPin patients who are adheringtofull doses of an appropriatethree-drugregimen that includes adiuretic(for2 months). After excludingpotential identifiable causes ofresistanthypertension (table9),With particular attention paid to diuretictypeand dosein relation to renal function, Consultation with a specialist should be considered .

Table (9): SomePotential Causes ofResistant Hypertension
• ImproperBPmeasurement • Volumeoverload and pseudo-tolerance - Excess sodium intake - Volume retention from kidneydisease - Inadequatediuretictherapy • Druginduced orothercauses - Non adherence to medication protocol - Inadequate doses(that does not cover 24 hours) - Inappropriate combinations of medications - Non-steroidal anti-inflammatories - Cocaine, amphetamines, other illicit drugs - Sympathomimetics (oral decongestant medications) - Oral contraceptives - Corticosteroids - Cyclosporine, tacrolimus - Erythropoietin - Licorice, including some chewing tobacco - Selected dietary supplements • Associated conditions - Obesity - Excess alcohol intake - White coat • Secondary hypertension

ReferralGuidelines for PHCCs

A hypertensivepatient should be referred to an appropriatespecialist underthe following conditions:

1. All hypertension emergencies cases should be referred to hospital
2. Iftherapeuticgoals, includingblood pressure control, havenot been reachedwithin six months.
3. For further assessment, screening,and management ofHMODor associated conditions, such as:

- Hemorrhage and exudates orpapilleodemaon fundoscopicexamination examination)
- Suspicion ofleft ventricular failure
- Suspicion of cerebrovascular accident ortransient ischemic attack
- Suspicion ofcoronaryevent
- Progressiveincreasein serum creatinineorrenal failure
- Hypertensiveencephalopathy
- (Anystage)peripheral vasculardisease
- Anypossibilityofsecondaryhypertension

Followupandmonitoring

Patients treated onlywith lifestylemodification should beseen after3-6 months. Aftertheir blood pressureis stabilized, theyshould beseenevery6-12 months to reinforcethenecessity and goals ofmaintaining thesemodifications oflifestyle.
Onceantihypertensivedrugtherapy isinitiated,mostpatientsshouldreturnforfollowupand adjustmentofmedicationsatapproximately monthlyintervalsuntiltheBPgoalisreached. Morefrequentvisitswillbenecessary forpatientswithstageIIhypertensionorwith complicating co-morbid conditions.

Monitoring parameters:

On each regular visit:

- Adherence to medication protocol
- Medication side effects
- Blood Pressure
- Weight and BMI
- Cardiac Examination
- Pulmonary Examination
- Optic Fundi
- Review of counseling on diet (especially salt intake), exercise, smoking

Every6months:

- Creatinine/ Urea Nitrogen
- Serum potassium
- ECG

Every12 months:

- A full fundoscopicexamination
- ECG – 12 lead
- Echocardiography
- Lipid profile
- Blood sugar

Patients who experienced a hypertensive emergency areat increased risk of cardiovascular and renal disease. Thorough investigation of potential underlying causes and assessment of HMOD is mandatory to avoid recurrent presentations with hypertensive emergencies.

Co-morbidities, such as heart failure, associated diseases such as diabetes, and the need for laboratory tests influence the frequency of visits. Other cardiovascular risk factors should be treated to their respective goals, and tobacco avoidance should be promoted vigorously. Low- dose aspirin therapy should be considered only when BP is controlled, because the risk of hemorrhagic stroke is increased in patients with uncontrolled hypertension.

Prevention of hypertension

High blood pressureis preventableand treatableand becausehypertensionis amajor risk factor forcardiovasculardisease. Manydeaths can beprevented ifhypertension is prevented from developingoris detected earlyand managedproperly.

Primaryprevention:

This refers toactions that can betaken priorto theonset ofhypertension through lifestylemodification.

Actions:

- Maintain normal bodyweight (BMI20-<25 kg/m)
- Avoid high salt diet(notadd salt to food eitherin cookingor at thetable,avoid cannedand processed foods)
- Engagein regularaerobic activity
- Limit alcohol consumption
- Consumediet rich in fresh fruits and vegetables,
- Replacesaturated fat withunsaturated fats.
- Stop smoking

Secondaryprevention:

This refers toactions which halt theprogress ofhypertension at its incipient stageand preventscomplications throughearlydetection and propermanagement.

Actions:

- Regularblood pressurescreening
- Lifestylemodification
- Earlyinitiation oftreatment and assuranceof compliance

Tertiaryprevention:

Refers to Measures usedlatein thestageofthediseaseto limit diseaseprogression andclinical disease complication afterovertclinical hypertension manifest

Actions:

- Lifestylemodifications
- Treatment ofhypertension and anytarget organ damagewith consideration of compliance, adherence,and concordance.
- Concomitant treatment ofdiseases developed secondaryto hypertension.

Health Education Messages on Hypertension

It is verycrucial that thepatient should understand what hypertension is, the factors that influenceit, thereforethetreatingphysician should haveaplanforthepatient education, stressingtheimportanceofthe followingpoints:

- Hypertensionrequires permanent, life-longtreatment it can be controlled,but not cured
- Hypertensionisfrequently asymptomatic and may progress silently (called the silent killer); therefore regular monitoring and management of BP is important.
- Hypertension is important risk factor for many cardiovascular diseases such as heart attack and stroke, which can be prevented by adequate control.
- Life style modification is an essential part of therapy which leads to better control of hypertension and reduces the no. and doses of required drugs.
- Hypertension is usually asymptomatic and requires treatment based only the blood pressure readings rather than symptoms such as headache or dizziness
- Stress can contribute to hypertension, but is not the most common cause; a (personal predisposition is the most common factor)Continually reinforce the dangers to smoking, not only in regard to hypertension, but to cardiac and lung health.
- Both systolic and diastolic BP are important, SBP may be more predictive of complication than DBP especially in elderly people.
- Encouraging selfmonitoring to give guide for management & not for deciding management.

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Hypertension

Prevention,Diagnosis,andTreatment

Annexes

Annex IEquipmentforBloodPressureMeasurement

A ***sphygmomomanometer***orblood pressuremeteris adevicewith an inflatable cuffused to measureblood pressure,consist ofinflation cuff,measuringunit (themanometer)tubeto connect thetwoand inflation bulb connected byatubeto the cuff.

Types ofsphygmomanometers:
1. ***Manometric***: includes both themercurycolumn typeand the aneroid, both manually inflated and requirestethoscope. Themercurytypeis veryaccurate and used for standardization ofothermeasuringdevices. Theaneroid may giveinaccurate readings ifnot properlycalibrated.

2. ***Digital types***: areeasyto usebut cangiveinaccurateresults, requireno stethoscope. Three types areavailable: *finger* and *wrist* devises, both arenot considered to beaccurate and the *upper arm* deviceswhicharetheleast susceptibleto position error.

Cuffsizeformeasuringblood pressure:

It is important to select thepropercuffsize forblood pressuremeasurement, too small cuffs givefalsehigh BP, and too large cuffsgivefalselow BPreading.The cuffbladderlength should be at least 80%ofthe arm circumference,and width should be at least 40%ofthe arm circumference (table3).

sizeofcuff

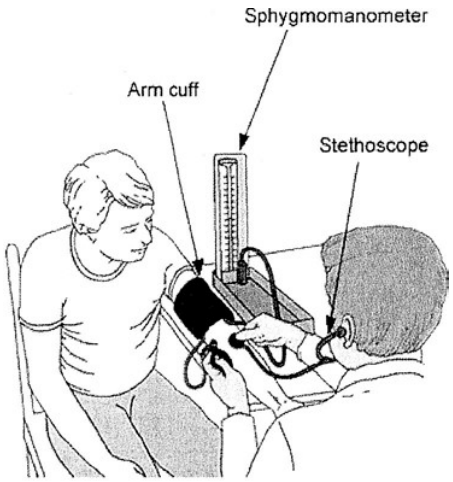
Midpoint Arm Circumference*** (cm)	Cuff Name	Bladder Width (cm)	Bladder Length (cm)
22-26	Small adult	10	24
27-34	Adult	13	30
35-44	Large adult	16	38
45-52	Thigh	20	42
*** Arm circumference as measured at the midpoint of the upper arm.			

MeasurementofBPatthe clinic/office:

1. Theauscultatorymethod ofBPmeasurement with aproperlycalibratedand

validated instrument should beused.

2. Persons should stop smokingand coffeedrinks for at least 30 min empty the bladder beforethe examination, and beseated quietlyforat least 5 minutes in a chair (ratherthan onan exam table), with feet on the floor, and arm supported at heart level (fig.1)



Propermeasurementofblood pressure

3. Measurement ofBPinthestandingposition is indicated periodically,especiallyin those at risk forpostural hypotension, diabetics and elderly.
4. An appropriate-sized cuffshould beused to ensure accuracy, rememberthat forobese patients appropriatecuffsizeshould beused; otherwise falsehighBPwill be recorded (table two).

5. Steps forproperbloodpressuremeasurement should be followed (table4). Systolic and diastolicblood pressureis accordinglyrecorded (table5)

5. At least two measurements should bemade andtakethe average. Both RT andLT armBP should bemeasured in theinitial visit, the arm with thehigherBPshould be considered for diagnosis ofhypertension and treatment decisions.

Clinicians should provideto patients, verballyand in writing, theirspecificBPnumbersandBPgoals.

Diagnosisof hypertensionisnotbasedonthe firstassessment.Bloodpressure measurement shouldbe takenthree separate timesone toseveralweeksfromthe initialassessmentto confirmthediagnosis.

Stepsforproperblood pressuremeasurementtechnique

- Expose theupperarm. Removeanytightor restrictiveclothingfromthearm
- Evaluate thepatient'sbareupperarmfor theappropriatesizecuff
- Placethecuffonthepatient'sbareupperarm,withtheloweredgeofthecuff2.5cmabove theantecubital fossa, with thecenterof thecuffbladderover thebrachialartery.
- Palpatebrachialarterypulse
- Inflate thecuffuntilpulsation disappears
- Deflate thecuff
- Thepointofdisappearance is theestimated systolicpressure
- Wait15-30seconds,thenplacethebellheadofthestethoscopeoverthebrachialarteryand inflate thebladder to 30 mmHgaboveestimated SBP
- Allowthecufftoslowlydeflateatarateof2-3mmHgpersecondwhilelisteningfor repetitivesounds
- Recordthepressureatwhichthefirstofatleasttworepetitivesoundsisheard.Thisisthe systolicblood pressure (phase1 sounds)
- Recordthepressureatwhichthelastregularsoundisheard.Thisisthediastolicblood pressure(phase5 sounds)
- Continue to listen duringfulldeflation to confirmdisappearanceof theheartsounds.

Auscultatory sounds inreading blood pressure

- **Phase1**thefirstappearanceoffaint,repetitive,cleartappingsoundsthatgradually increase in intensityforat least two consecutivebeats is thesystolicblood pressure.
- **Phase2**Abriefperiodmayfollowduringwhichthesoundssoftenandacquireas wishing quality
- **Auscultatory gap** in somepatientssoundsmaydisappearaltogether forashort time
- **Phase3** thereturn ofsharpersounds,maybeeven sharper than phase1 sounds.
- **Phase4** thedistinct, abruptmufflingsounds, which becomesoftand blowingin quality
- **Phase5** Thepointatwhich allsounds finallydisappearcompletelyis thediastolicpressure

Inaccurateblood pressuremeasurement:

Accurateblood measurement is essential fordiagnosis and treatment ofhypertensivepatients. The followingpoints need to be consideredforinaccuracy:

1. **Inaccuratemanometerorimproper cuffsize**.
   Themanometershould be calibrated at regularbasis and thepropercuffsizeshould be selected.
2. **Inaccuracy results from the examiner**: cuffpressureis releasedtoo fast, terminal digit bias (the examinertendsto round offthemeasured valueso as to end in 0or5), examiner bias and the examinerskills.
3. **Inaccuracy results frominter-armdifference**:
   Pressuredifferenceof>10 mm Hgbetween the arms ofhypertensivepatient maybepresent in hypertensivepatient, in such a casethehigherBPreadingshould beselected fordiagnosis and treatment.
4. **Inaccuracy fromphysiological variation**:
   Emotional situation ofthepatient(stress orfear)duringthemeasurement, white coat hypertension, thepatientactivitypriorto measurement, position duringmeasurement(sitting, standing, seating)and ingestion ofcertain substances (caffeine, alcohol , nicotineor sympathomimeticdrugs likenasaldrops)

Ambulatory BP monitoring:

Ambulatoryblood pressuremeasurement is amethod ofobtaininginformation about blood pressurein apatient outsidethe clinical setting.

Anxiety, discomfort andunfamiliarsurroundingsat the clinic can lead to transient risein BP. Sphygmomanometry, especiallywhen performed byphysicians,can causeunrepresentative surgein BP called "**white coathypertension**" and up to 20%ofpatients with apparent hypertension in the clinic, tend to havenormalBPrecordings byautomateddevises measured at home.

Ambulatory BPmeasurement devises areportableautomated instruments that can beheld by thepatient forthewhole24 hours, recordingtheBPautomaticallyat specified timeintervals duringtheactual patientdaylife.

Ambulatory BPrecordings arebetter correlated with thetarget organ damagedueto hypertension than clinicreadings ,yet lowerreadingthan theclinic recordings should be considered forboth diagnosis and treatment decisions(clinic readingshould be>140/85 , ambulatory>130/80).In most individuals, BPdecreases by10 to 20 percent duringthe night; thosein whom such reductions arenot present areat increased risk for cardiovascular events.

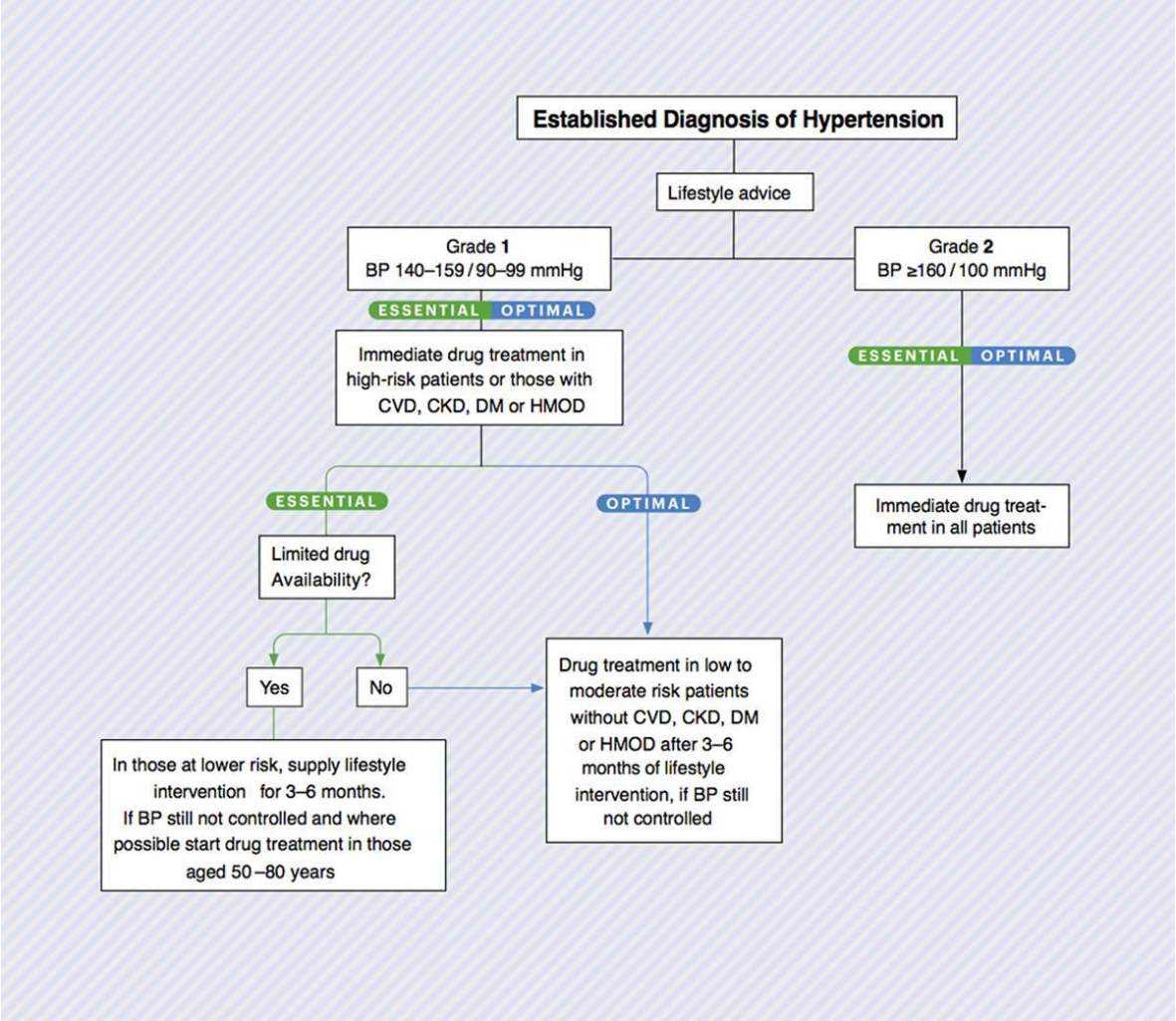
Indicationsforambulatory blood pressuremonitoring:

1. White-coat"hypertension in the absenceoftarget organ damage.
2. Drugresistancehypertension.
3. Hypotensivesymptoms with antihypertensivemedication
4. Episodicorlabilehypertension and autonomicdysfunction

Self-measurementofblood pressure:

Blood pressureself-measurements maybenefit patients byprovidinginformation on response to antihypertensivemedication, improvingpatientadherencewith therapy,and in evaluating white-coat hypertension.
Persons with an averageBPmorethan 135/85 mmHgmeasuredat homearegenerally considered to behypertensive. Homemeasurement devices which areofvariablequality semi-automated devicesshould be checked regularlyfor accuracy.

Annex II Pharmacological treatment of hypertension: general scheme



International society of hypertension global hypertension practice guidelines(2020)

Annex III ALifestyle Modifications

Salt reduction	There is strong evidence for a relationship between high salt intake and increased blood pressure. Reduce salt added when preparing foods, and at the table. Avoid or limit consumption of high salt foods such as soy sauce, fast foods and processed food including breads and cereals high in salt
Healthy diet	Eating a diet that is rich in whole grains, fruits, vegetables, polyunsaturated fats and dairy products and reducing food high in sugar, saturated fat and trans fats, such as the DASH diet (http://www.dashforhealth.com). Increase intake of vegetables high in nitrates known to reduce BP, such as leafy vegetables and beetroot. Other beneficial foods and nutrients include those high in magnesium, calcium and potassium such as avocados, nuts, seeds, legumes and tufo
Healthy drinks	Moderate consumption of coffee, green and black tea. Other beverages that can be beneficial include karkadé (hibiscus) tea, pomegranate juice, beetroot juice and cocoa.
Moderation of Alcohol consumption	Positive linear association exists between alcohol consumption, blood pressure, the prevalence of hypertension, and CVD risk. The recommended daily limit for alcohol consumptions is 2 standard drinks for men and 1.5 for women (10 g alcohol/standard drink). Avoid binge drinking.
Weight reduction	Body weight control is indicated to avoid obesity. Particularly abdominal obesity should be managed. Ethnic-specific cut-offs for BMI and waist circumference should be used. Alternatively, a waist-to-height ratio <0.5 is recommended for all populations.
Smoking cessation	Smoking is a major risk factor for CVD, COPD and cancer. Smoking cessation and referral to smoking cessation programs are advised.
Regular Physical activity	Studies suggest that regular aerobic and resistance exercise may be beneficial for both the prevention and treatment of hypertensio Moderate intensity aerobic exercise (walking, jogging, cycling, yoga, or swimming) for 30 minutes on 5–7 days per week or HIIT (high intensity interval training) which involves alternating short bursts of intense activity with subsequent recovery periods of lighter activity. Strength training also can help reduce blood pressure. Performance of resistance/strength exercises on 2–3 days per week
Reduce stress and mindfulness	Chronic stress has been associated to high blood pressure later in life. Although more research is needed to determine the effects of chronic stress on blood pressure, randomized clinical trials examining the effects of transcendental meditation/mindfulness on blood pressure suggest that this practice lowers blood pressure. Stress should be reduced and mindfulness or meditation introduced into the daily routine.
Complimentary alternative or traditional medicines	Large proportions of hypertensive patients use complementary, alternative or traditional medicines (in regions such as Africa and China) yet large-scale and appropriate clinical trials are required to evaluate the efficacy and safety of these medicines. Thus, use of such treatment is not yet supported.
Reduce exposure to air pollution and cold temperature	Evidence from studies support a negative effect of air pollution on blood pressure in the long-term.

*(2020)International Society Of Hypertension Global Hypertension Practice Guidelines with modification

Annex III BDASH: Dietary Approaches to Stop Hypertension

Food	Servings per day	Serving Size
Grains	5-6 daily servings	½ cup cooked rice or pasta; 1 slice whole grain bread
Vegetables	4-5 daily servings	1 cup raw vegetables or fruit ½ cup cooked vegetables or fruit
Fruits	2-3 daily servings	1 cup raw fruit 1 medium banana 1 orange, ½ grapefruit
Low-fat or fat-free dairy products	2-3 daily servings	1 cup of low-fat milk or yoghurt
Lean meat, poultry and fish	2 daily servings	90 grams cooked lean meat 90 grams cooked chicken
Nuts, seeds, and dry beans/lentils	1-2 servings per day	15 grams of nuts ¼ cup cooked beans, peas, or lentils, humus
Fats and oils	2-3 daily servings.	1 teaspoon olive oil
Sweets - limited	Less than 1 serving per day	1 small piece of pastry or sweet

Note: DASH diet adapted slightly for approximately 1600 Kcal/day, and to control simple sugars. See also “Diet for Diabetes and Metabolic Syndrome” for general suggestionshttp://www.nhlbi.nih.gov/health/public/heart/hbp/dash/new_dash.pdf
For dietary guideline see WHO EMRO publication Promoting a healthy diet for the WHO Eastern Mediterranean Region: user-friendly guide.

Annex IV Oral Antihypertensive Drugs

Class	Drug	Usual Dose Renge (mg/d)	Daily Frequency	
Primary agents				
Thiazide or thaizide – type diuretics	Chlorthalidone	12.5-25	1	• Chlorthalidone is preferred on the basis of prolonged half-life and proven trial reduction of CVD. • Monitor for hyponatremia and hypokalemia, uric acid and calcium levels. • Use with caution in patients with history of acute gout unless patient is on uric acid–lowering therapy.
	Hydrochlorothiazide	25-50	1	
	Indapamide	1.25-2.5	1	
	Metolazone	2.5-5	1	
ACE inhibitors	Benazepril	10-40	1 or 2	• Do not use in combination with ARBs or direct renin inhibitor. • There is an increased risk of hyperkalemia, especially in patients with CKD or in those on K supplements or K-sparing drugs. • There is a risk of acute renal failure in patients with severe bilateral enal artery stenosis. • Do not use if patient has history of angioedema with ACE inhibitors. • Avoid in pregnancy.
	Captopril	12.5-150	2 or 3	
	Enalapril	5-40	1 or 2	
	Fosinopril	10-40	1	
	Lisinopril	10-40	1	
	Moexipril	7.5-30	1 or 2	
	Perindopril	4-16	1	
	Quinapril	10-80	1 or 2	
	Ramipril	2.5-20	1 or 2	
	Trandolapril	1-4	1	
ARBS	Azilsartan	40-80	1	• Do not use in combination with ACE inhibitors or direct renin inhibitor. • There is an increased risk of hyperkalemia in CKD or in those on K supplements or K-sparing drugs. • There is a risk of acute renal failure in patients with severe bilateral renal artery stenosis. • Do not use if patient has history of angioedema with ARBs. Patients with a history of angioedema with an ACE inhibitor • can receive an ARB beginning 6 weeks after ACE inhibitor is discontinued. • Avoid in pregnancy.
	Candesartan	8-32	1	
	Eprosartan	600-800	1 or 2	
	Irbesartan	150-300	1	
	Losartan	50-100	1 or2	
	Olmesartan	20-40	1	
	Telmisartan	20-80	1	
	Valsartan	80-320	1	
CCB-dihydropyridines	Amlodipine	2.5-10	1	• Avoid use in patients with HFrEF; amlodipine or felodipine may be used if required. • They are associated with dose-related pedal edema, which is more common in women thanmen.
	Felodipine	2.5-10	1	
	Isradipine	5-10	2	
	Nicardipine SR	60-120	2	
	Nifidipine LA	30-90	1	
	Nisoldipine	17-34	1	
CCB nondihyropyridines	Ditiazem ER	120-360	1	• Avoid routine use with beta blockers because of increased risk of • bradycardia and heart block. • Do not use in patients with HFrEF. • There are drug interactions with diltiazem and verapamil (CYP3A4 • major substrate and moderate inhibitor).
	Verapamil IR	120-360	3	
	Verapamil SR	120-360	1 or 2	
	Verapamil-delayed onset ER	100-300	1(in the evening)	
Secondary agents				
Diuretics-Loop	Bumetanide	0.5-2	2	• These are preferred diuretics in patients with symptomatic HF. They are preferred over thiazides in patients with moderate-to-severe CKD (e.g., GFR< 30 ml per min)
	Furosemide	20-80	2	
	Torsemide	5-10	1	
Diuretics-potassium sparing	Amiloride	5-10	1 or 2	• These are monotherapy agents and minimally effective antihypertensive agents. • Combination therapy of potassium-sparing diuretic with a thiazide can be considered in patients with hypokalemia on thiazide monotherapy. n • Avoid in patients with significant CKD (e.g., GFR <45 mL/min).
	Triamterene	50-100	1 or 2	

Annex IV Oral Antihypertensive Drugs (continued)

Class	Drug	Usual Dose Renge (mg/d)	Daily Frequency	
Diuretics—aldosterone antagonists	Eplerenone	50–100	1 or 2	These are preferred agents in primary aldosteronism and resistant hypertension. • Spironolactone is associated with greater risk of gynecomastia and impotence as compared with eplerenone. • This is common add-on therapy in resistant hypertension. • Avoid use with K supplements, other K-sparing diuretics, or significant renal dysfunction. • Eplerenone often requires twice-daily dosing for adequate BP lowering.
	Spironolactone	25-100	1	
Betablockers-cardioselective	Atenolol	25-100	2	• Beta blockers are not recommended as first-line agents unless the patient has IHD or HF. • These are preferred in patients with bronchospastic airway disease requiring a beta blocker. • Bisoprolol and metoprolol succinate are preferred in patients with HFrEF. • Avoid abrupt cessation
	Betaxolol	5-20	1	
	Bisoprolol	2.5-10	1	
	Metoprololtartarate	100-200	2	
	Metoprolol succinate	50-200	1	
Betablockers-Cardioselective and vasodilatore	Nebivolol	5-40	1	• Nebivolol induces nitric oxide–induced vasodilation. • Avoid abrupt cessation
Beta blockers noncardioselective	Nadolol	40 -120	1	• Avoid in patients with reactive airways disease. • Avoid abrupt cessation.
	Propranolol IR	80 -160	2	
	Propranolol LA	80 - 160	1	
Beta blockers - intirinsic sympathomimetic activity	Acetabutolol	200 -800	2	• Generally avoid, especially in patients with IHD or HF. • Avoid abrupt cessation .
	Penbutolol	10 -40	1	
	Pindolol	10 -60	2	
Beta blockers—combined alpha- and beta-receptor	Carvedilol	12.5 -50	2	• Carvedilol is preferred in patients with HFrEF. • Avoid abrupt cessation.
	Carvedilol phosphate	20 -80	1	
	Labetolol	200 -800	2	
Direct renin inhibitor	Aliskiren	150 -300	1	• Do not use in combination with ACE inhibitors or ARBs. • Aliskiren is very long acting. • There is an increased risk of hyperkalemia in CKD or in those on K supplements or K-sparing drugs. • Aliskiren may cause acute renal failure in patients with severe bilateral renal artery stenosis. • Avoid in pregnancy
Alpha 1-blockers	Doxazosin	1 - 16	1	These are associated with orthostatic hypotension, especially in older adults. • They may be considered as second-line agent in patients with concomitant BPH
	Prazosin	2 -20	2 or 3	
	Tetrazosin	1 -20	1 or 2	
Central alpha2-agonist and other centrally acting drugs	Clonidine oral	0.1 -0.8	2	• Central alpha2-agonist and other centrally acting drugs • These are generally reserved • These are generally reserved as last-line because of significant CNS adverse effects, especially in older adults. • Avoid abrupt discontinuation of clonidine, which may induce hypertensive crisis; clonidine must be tapered to avoid rebound hypertension.
	Clonidine patch	0,1 -0.3	1 weekly	
	Methyl dopa	250 -1000	2	
	Guafacine	0.5 -2	1	
Direct vasodilators	Hydralazine	100 -200	2 or 3	• These are associated with sodium and water retention and reflex tachycardia; use with a diuretic and beta blocker • Hydralazine is associated with drug-induced lupus-like syndrome at higher doses. • Minoxidil is associated with hirsutism and requires a loop diuretic. • Minoxidil can induce pericardial effusion
	Minoxidil	5 -100	1-3	